ON A REGULAR SCHEME A COHERENT MODULE IS COHEN-MACAULAY WITH FULL SUPPORT IF AND ONLY IF IT IS FINITE LOCALLY FREE OF POSITIVE RANK.

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3109-3117 of sources/stacks-coherent/pinned-original-source.tex; proof span is lines 3119-3130, the complete author proof (10 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Pool stacks-coherent backfill wave 33h (tail batch, final screening of ranks 86-118).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. See why-field; every pointer named here occurs verbatim in the retained statement/proof text.

Source reference. Original lemma 00NF: sources/stacks-coherent/pinned-original-source.tex, source lines 3130–3130. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00NT: sources/stacks-coherent/pinned-original-source.tex, source lines 3130–3130. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00NQ: sources/stacks-coherent/pinned-original-source.tex, source lines 3130–3130. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let X be a regular scheme. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. The following are equivalent*

(1) *$\mathcal{F}$ is Cohen-Macaulay and $Supp(\mathcal{F}) = X$,*
(2) *$\mathcal{F}$ is finite locally free of rank > 0.*

Proof. Let $x \in X$. If (2) holds, then $\mathcal{F}_x$ is a free $\mathcal{O}_{X,x}$-module of rank > 0. Hence $\text{depth}(\mathcal{F}_x) = \dim(\mathcal{O}_{X,x})$ because a regular local ring is Cohen-Macaulay (Algebra, Lemma 00NQ). Conversely, if (1) holds, then $\mathcal{F}_x$ is a maximal Cohen-Macaulay module over $\mathcal{O}_{X,x}$ (Algebra, Definition 00NF). Hence $\mathcal{F}_x$ is free by Algebra, Lemma 00NT. □

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